
Applied Artificial Intelligence

Master1

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Chapter 1: Introduction to Artificial Intelligence

What is Artificial Intelligence?

Artificial Intelligence (AI) is a field of computer science and engineering that focuses on **designing systems capable of performing tasks** that normally require human intelligence, such as learning, reasoning, perception, and decision-making. In engineering applications, **AI systems analyze data and produce predictions or decisions** to assist or automate complex processes.

Unlike traditional programming, where system behavior is fully defined by explicit rules written by a programmer, AI systems learn their behavior from data. This learning-based approach allows AI to handle uncertainty, variability, and complexity, which are common in real-world systems.

EX:

In classical programming, an engineer writes explicit rules such as: **“If temperature > 80°C, stop the machine.”** In Artificial Intelligence, **the system learns from historical data** how temperature, vibration, and pressure together indicate a fault, without explicitly programming all possible rules.

□ In practice, Artificial Intelligence transforms raw data (signals, measurements, images, or text) into useful information. The output of an AI system can be a classification, a prediction, a detection, or a decision, making AI a powerful tool for electronics, automatic control, and industrial applications.

□ **Data → AI model → prediction / decision.**

□ Most AI systems used today are *narrow AI*, meaning they are designed to solve a specific task efficiently rather than to imitate general human intelligence.

□ **EX:** a chatbot designed for customer support: it can answer predefined types of questions but cannot reason outside its trained domain.

❑ **Historical Evolution of Artificial Intelligence**

❑ **1950s–1960s: Birth of Artificial Intelligence**

The term *Artificial Intelligence* was introduced by John McCarthy in 1954. During this period, two main approaches developed in parallel: numerical computation (scientific computing) and symbolic computation (formal reasoning). Early AI focused on exhaustively describing simple and general problems, relying on computer memory and computational power to generate and compare many possible solutions. This approach quickly faced the problem of combinatorial explosion.

❑ **1970s: Expert Systems**

The 1970s marked the development of expert systems, designed to solve very specific problems by representing specialized knowledge from a given domain. Intelligence became associated with decision-making in complex situations involving uncertainty, incomplete, or contradictory information. The notion of *expertise* replaced the earlier focus on simple and strictly formal knowledge.

❑ **1980s–1990s: Industrial Expansion**

During the 1980s and 1990s, AI experienced significant industrial adoption. Japan launched the Fifth Generation Computer Systems project, aiming to develop intelligent machines. At the same time, logical programming languages such as PROLOG, based on logical inference, became widely used and contributed to the spread of AI technologies.

❑ **Today: Modern Artificial Intelligence**

Today, Artificial Intelligence is driven by Machine Learning and Deep Learning techniques and is applied in numerous fields such as autonomous vehicles, intelligent systems, data analysis, and industrial automation. AI has become an integral part of many scientific, technological, and engineering domains.

Why Artificial Intelligence Matters Today

Artificial Intelligence has become essential because modern engineering systems generate large amounts of data through sensors, machines, and digital platforms. Classical rule-based and analytical methods often struggle to exploit this data efficiently, especially when systems are complex, uncertain, or non-linear. AI provides data-driven techniques capable of extracting useful information and patterns from such data.

From an engineering point of view, AI enables better decision-making, optimization, and automation. It is widely used for tasks such as fault detection, predictive maintenance, image inspection, signal analysis, and system monitoring.

Thanks to advances in computing power and open-source tools, AI techniques are now accessible and practical for engineers.

In a traditional approach, an engineer may define fixed rules such as stopping a machine when vibration or temperature exceeds a predefined threshold. This method is limited because it does not adapt to different operating conditions and may detect faults too late.

Instead of defining fixed thresholds, the system learns what ‘normal’ behavior looks like and detects deviations automatically.

FIELDS OF APPLICATION OF ARTIFICIAL INTELLIGENCE



Machine Learning and Deep Learning

- Machine Learning (ML) is a core approach in Artificial Intelligence that enables systems to automatically learn from data without being explicitly programmed with fixed rules. In Machine Learning, an algorithm analyzes examples (data) to build a model that captures relationships between inputs and outputs. Once trained, this model can be used to make predictions or decisions on new, unseen data. Machine Learning is widely used in engineering for tasks such as fault detection, signal classification, system modeling, and performance prediction.
- Machine Learning algorithms are generally categorized into different types. In **supervised learning**, the algorithm learns from labeled data, where the correct output is known (for example, classifying signals as normal or faulty). In **unsupervised learning**, the algorithm discovers patterns or structures in unlabeled data, such as grouping similar signals using clustering. A third category, **reinforcement learning**, focuses on learning optimal actions through interaction with an environment and is commonly used in control and robotics.

Machine Learning (ML) Example

Predictive Maintenance in Industry

An industrial system collects sensor data such as vibration, temperature, and pressure during machine operation. Using Machine Learning algorithms (e.g., decision trees or support vector machines), the system is trained to classify machine states as *normal* or *faulty*. The model relies on features extracted by the engineer and helps schedule maintenance before failures occur.

Deep Learning (DL) is a specialized branch of Machine Learning based on artificial neural networks composed of multiple hidden layers. These deep architectures allow the model to automatically learn hierarchical features from raw data, reducing the need for manual feature engineering. Deep Learning has shown outstanding performance in problems involving images, audio signals, and text, where traditional Machine Learning methods often struggle.

In practice, Deep Learning requires large datasets and significant computational resources, but it offers high accuracy and flexibility. While Machine Learning models are often simpler and easier to interpret, Deep Learning models are more powerful for complex tasks. For this reason, engineers typically choose between ML and DL depending on the problem complexity, available data, and computational constraints.

Deep Learning (DL) Example

Image Recognition in Autonomous Driving

Cameras mounted on a vehicle continuously capture images of the environment. A Deep Learning model, typically a convolutional neural network (CNN), processes raw images to detect objects such as pedestrians, traffic signs, and other vehicles. The model automatically learns visual features from the data and supports real-time driving decisions.

Aspect	Machine Learning (ML)	Deep Learning (DL)
Data type	Structured data	Large, complex data
Feature extraction	Done by the engineer	Learned automatically
Model complexity	Relatively simple	Highly complex (deep networks)
Data requirement	Small to medium datasets	Large datasets
Computation	Low to moderate	High (GPU often needed)
Interpretability	Easier to interpret	Harder to interpret
Typical examples	Fault detection, prediction	Image, speech, text analysis

Limitations and Ethical Issues of Artificial Intelligence

Examples for Limitations of AI

Example 1 – Poor data quality

An AI system trained to detect defects in electronic components may fail if the training data contains noisy sensor measurements or incomplete records. As a result, the model may miss real faults or generate false alarms.

Example 2 – Biased data

If an AI model is trained using data collected only from a specific type of machine or operating condition, it may perform poorly when applied to a different machine or environment.

Example 3 – Black-box models

A deep neural network may correctly classify a system failure, but engineers cannot easily explain *why* the decision was made. This makes troubleshooting and validation difficult in industrial systems.

Examples for Ethical Issues

Example 4 – Discrimination due to bias

An AI system used for job candidate selection may unfairly favor or disadvantage certain groups if historical training data contains biased hiring decisions.

Example 5 – Safety-critical errors

In autonomous driving, a wrong object detection by an AI system could lead to incorrect braking or steering, potentially causing accidents.

Example 6 – Lack of responsibility

If an AI system makes a wrong decision in a medical or industrial context, the responsibility lies with the human designers and operators, not the AI system itself.

Examples for Good Practices

Example 7 – Human-in-the-loop

In industrial fault detection, AI predictions are reviewed by engineers before maintenance decisions are made.

Example 8 – Decision support, not replacement

In healthcare or safety monitoring, AI assists experts by highlighting risks, but final decisions are taken by humans.